

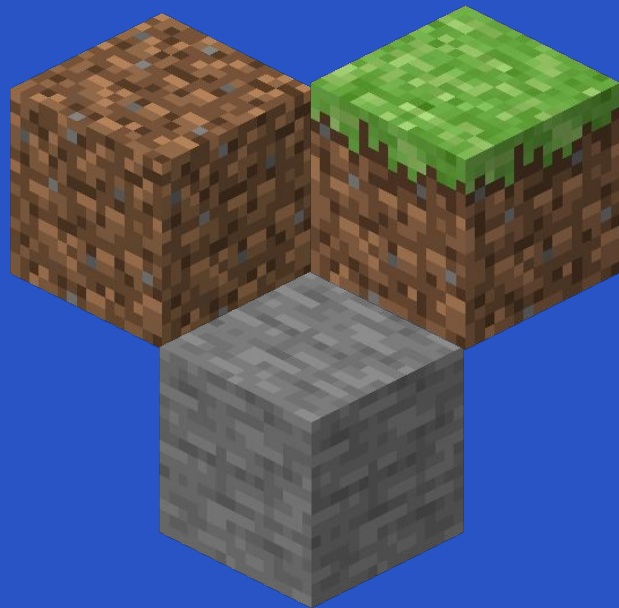
Minecraft Variational Autoencoder

Final Presentation

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Advisor: Dr. Shamir

CIS 598 Research Project
Spring 2026



Generative AI for Images & Text



ChatGPT

Gemini



Sora



Stable Diffusion



DALL-E

Research Opportunity: Machine Learning with Minecraft

Relatively little research:

- Little academic interest
- Few public datasets
- Difficult data to process

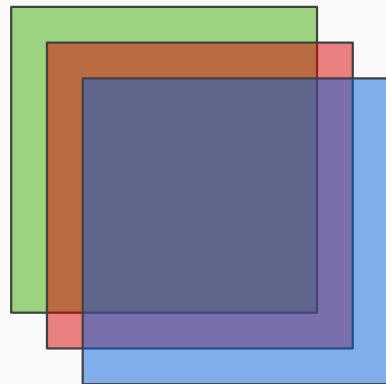
Minecraft

3D Discrete



Images

2D Continuous



WorldGAN (2021): Block2Vec

- Generative Adversarial Network (GAN)
- Adapted Word2Vec from NLP (Block2Vec)

Token Embedding Vectors

0: air := [0.22, -1.22, 3.14, ...]

1: stair  := [1.02, -3.42, 0.02, ...]

2: dirt  := [0.21, -2.62, -0.13, ...]

3: grass  := [-0.64, 1.41, 2.02, ...]

4: anvil  := [0.02, -0.32, 3.02, ...]

5: stone  := [1.10, 4.50, 1.12, ...]

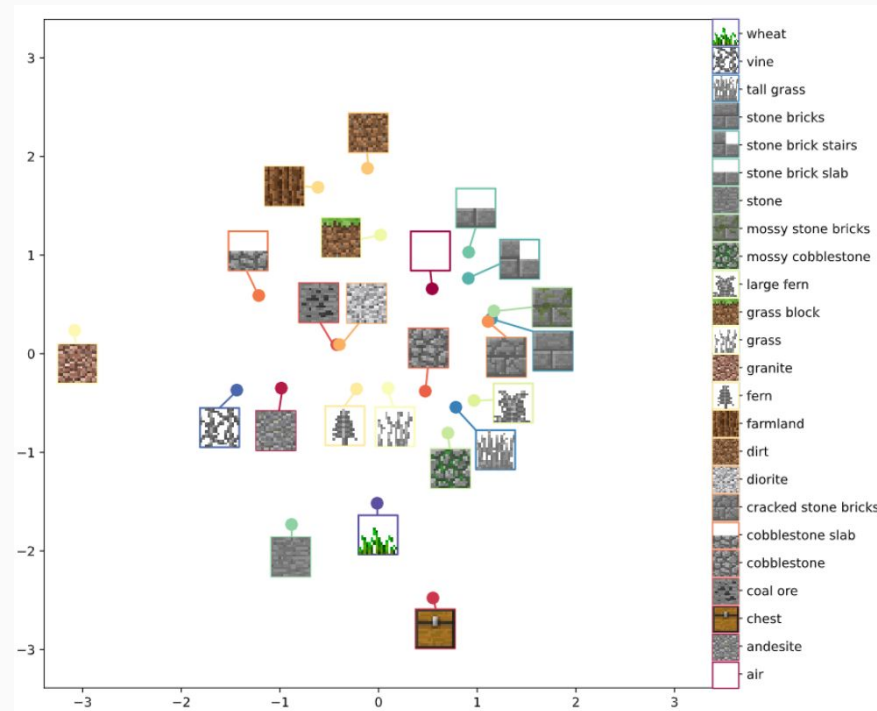


Image Source: WorldGAN(2021)

Interactive Latent Variable Evolution for the Generation of Minecraft Structures (2023)

- **One GAN for generating block/air voxels**
- **Another GAN for filling block voxels w/ block types**
- **Small structure dataset**

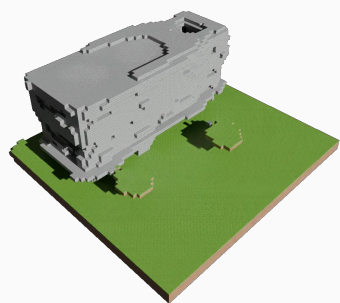
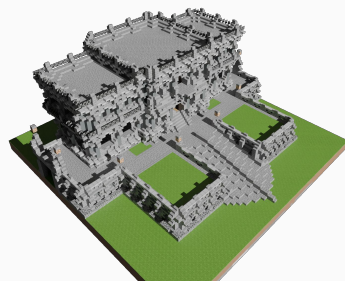
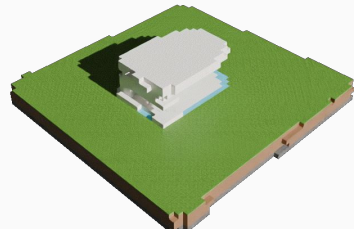
Text2MC (2024)

- VAE attempts to compress then reconstruct data
- Created VAE architecture for 3D token data
- Trained the model on ~10K Minecraft Builds
- Block2Vec from WorldGAN (2021)
- Block/Air and Block Type separation from ILVE (2023)

Original Build

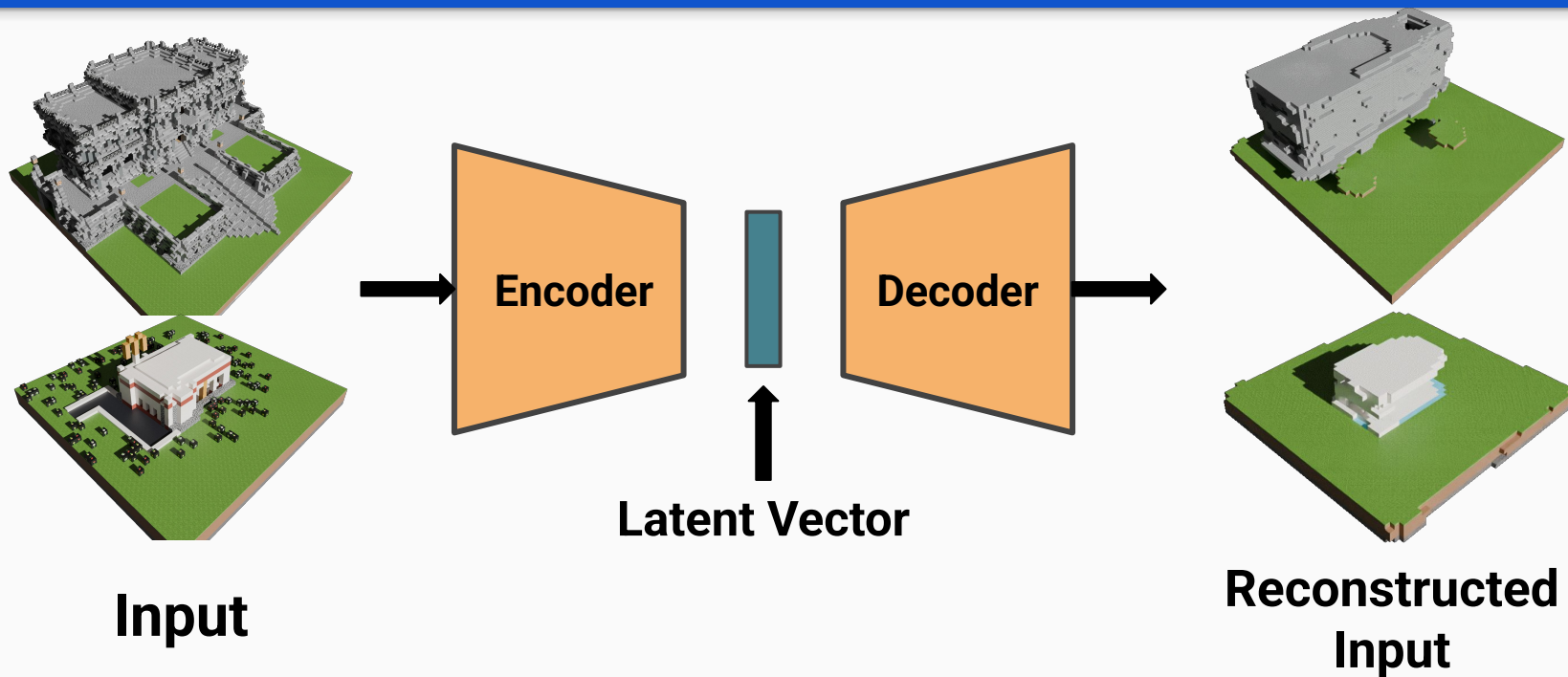


Reconstruction



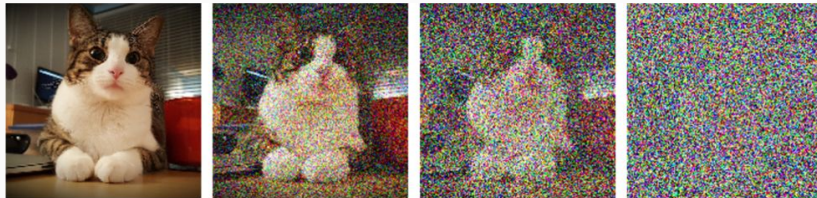
S. Comino et al., "Text2mc: Implementing Generative AI and Natural Language Processing In Minecraft," Github,
<https://github.com/shauncomino/text2mc-dataprocessor>.

What is Variational Autoencoder (VAE)?



VAE Use Cases

Text-Prompted Denoising Diffusion



Cluster Similar Builds

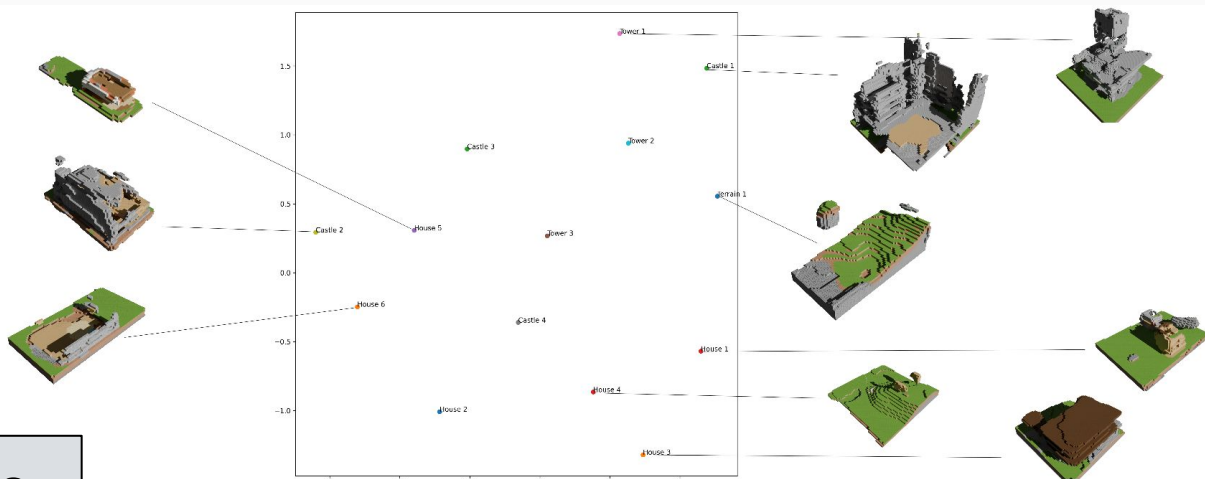
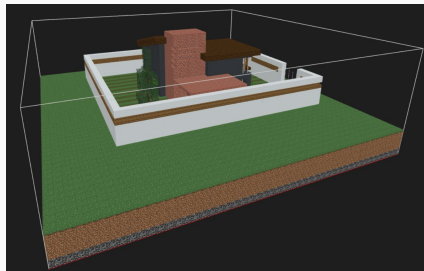


Image Source: Text2MC

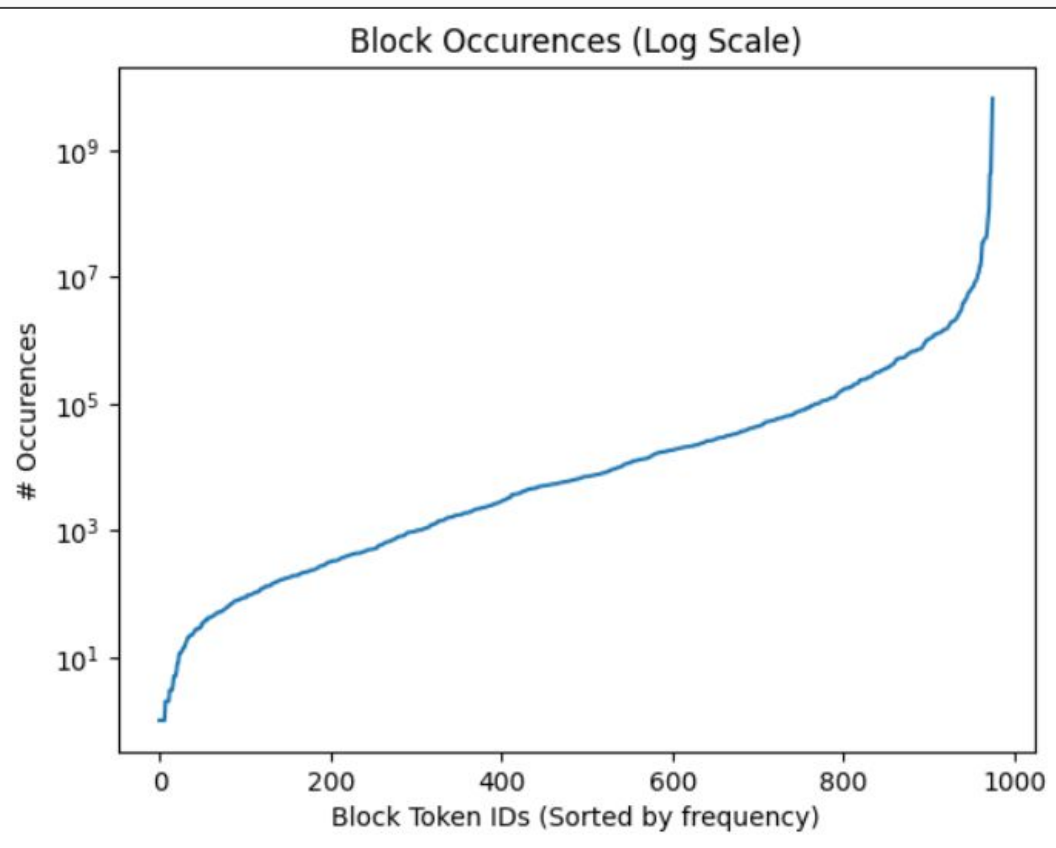
Dataset

- Sourced from Text2MC project.
- ~31,000 Minecraft Schematics.
- Different Sizes & Formats

Sample Examples



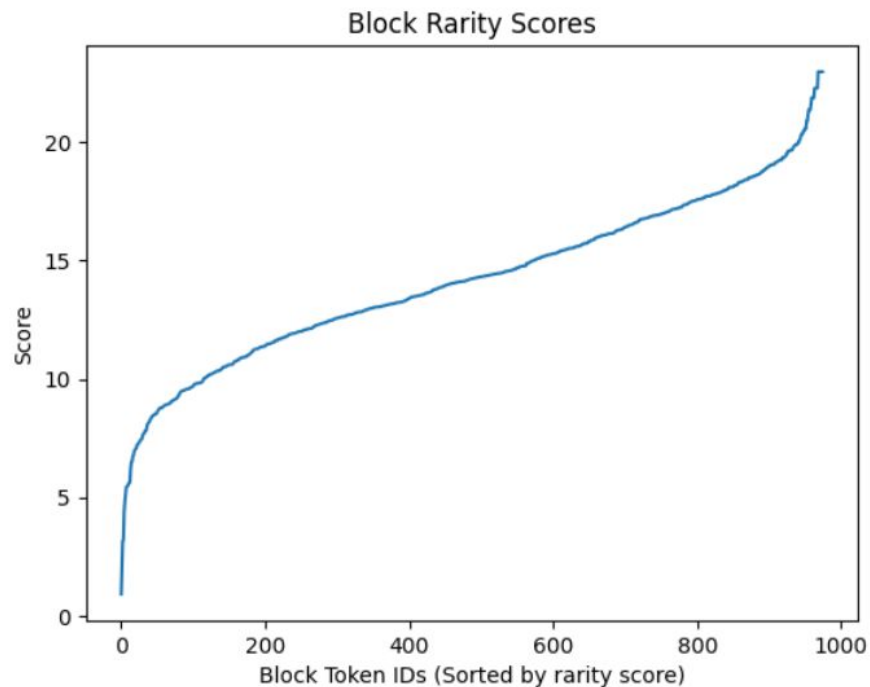
Block Type Distribution Across Dataset



Mask low-occurring blocks with air



Block Type “Density” Calculation



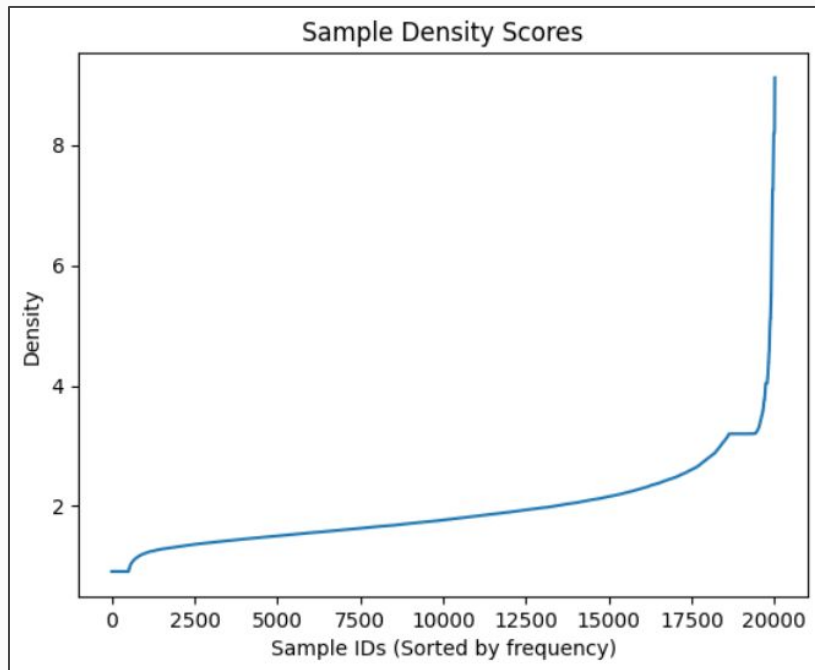
$$f(n) = \log(1 + (N / C_n))$$

n : a block token

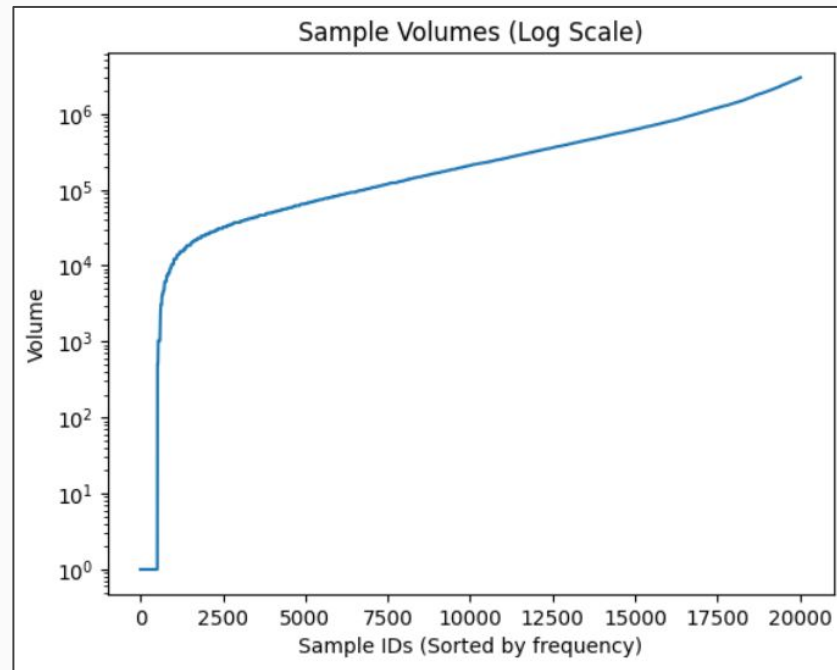
C_n : Count of block token across all samples

N : Sum of all sample volumes

Sample Volume & Density Distribution



Remove low / high density samples



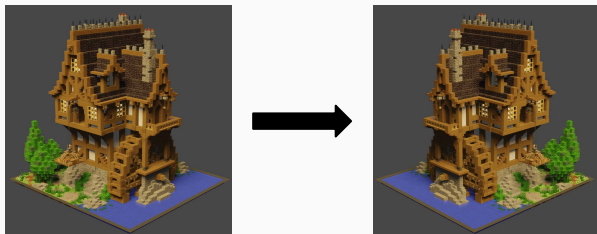
Remove low volume samples

Hypothesis & Research Questions

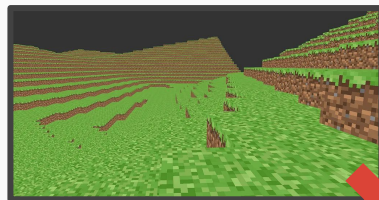
Research Question:

When retraining the Text2MC VAE, how will the following affect Reconstruction Quality?

1. Increasing Size of the Dataset
2. Weighted ROI Sampling
3. Random Reflection & Rotation Transformations



Reflection Transformation



Sample areas with structures

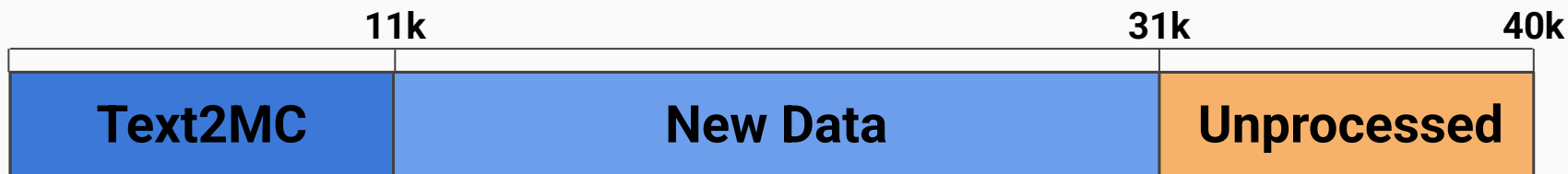
Hypothesis:

All three factors will have a positive effect on the Reconstruction Quality.

Independent Variables

Dataset Size

- Text2MC dataset processed 11K of 40K schematics.
- New dataset processed 20K more, excluded volumes > 3M.
- ~25GB → ~50GB



Prediction:

Increasing dataset size will improve performance, but have diminishing returns.

Independent Variables

Geometric Transformations

- Reflection & Rotations (excluding Y-axis)
- Apply the transformation to the 3D matrix AND the tokens (orientation)



64 x 32 x 48

90° Rotation
X-Axis Reflection



48 x 32 x 64

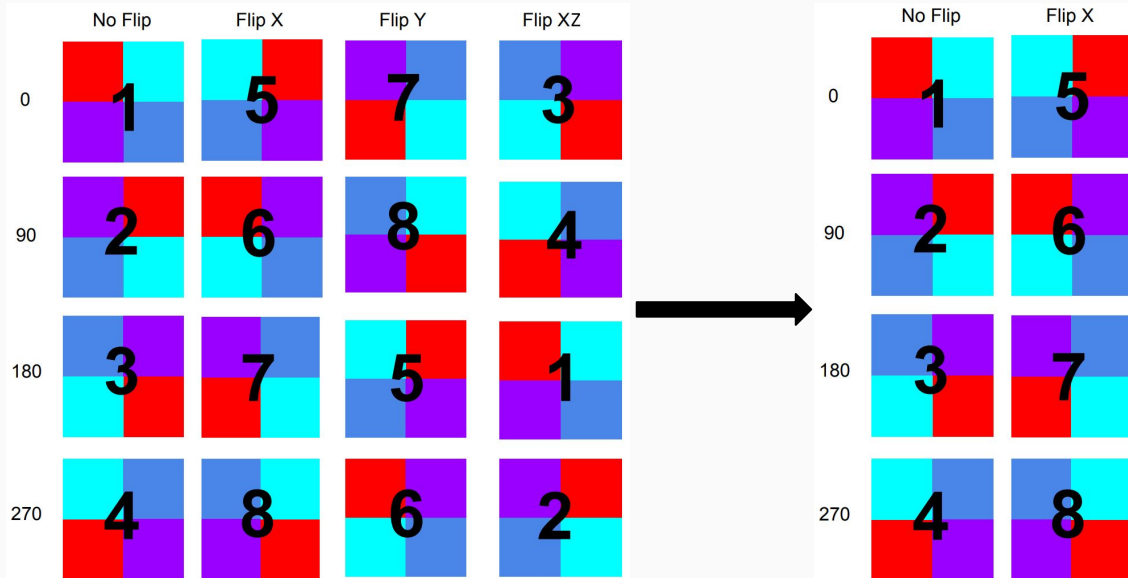
Prediction:

Applying random transformations will improve performance by preventing overfitting.

Independent Variables

Geometric Transformations

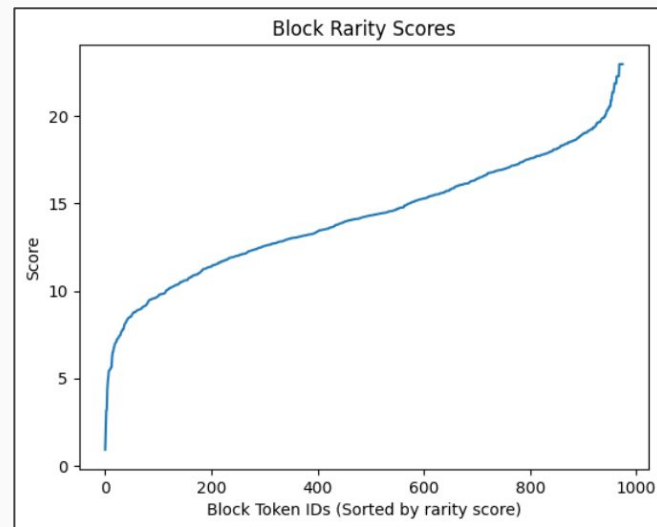
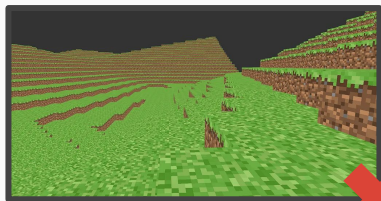
8 Unique Transform Combinations



Independent Variables

Weighted ROI Sampling

- Block “rarity” scores → 3D heatmap.
- Terrain blocks are more common.
- Weighted Random selection using heatmap.

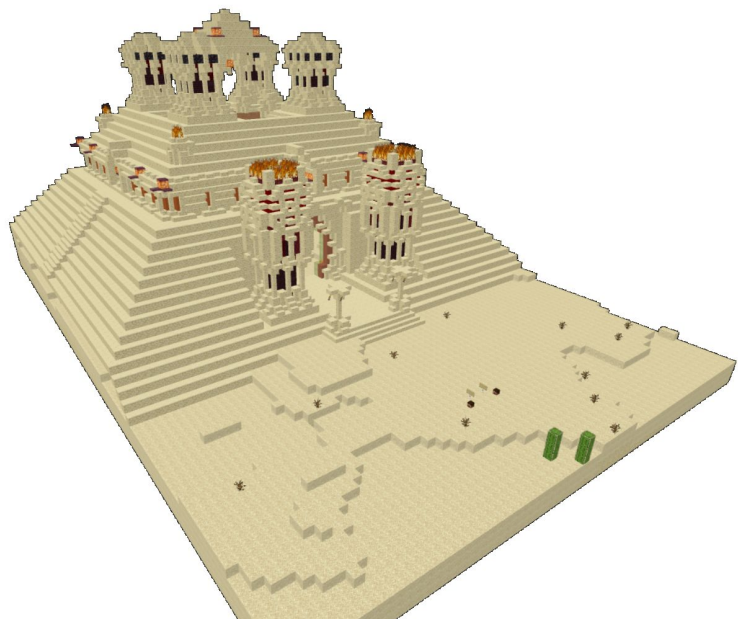


Prediction:

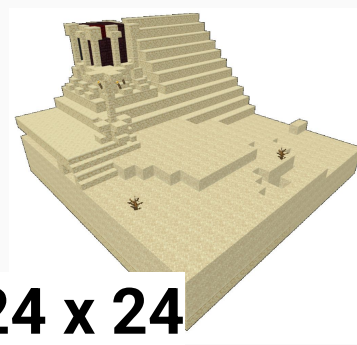
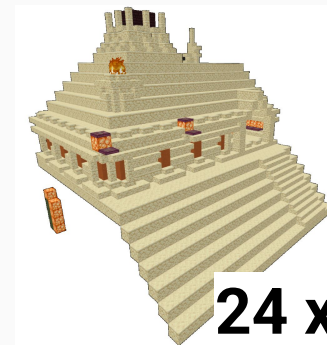
Applying weighted ROI will improve performance by training on less terrain & more structures.

Independent Variables

Weighted ROI Sampling



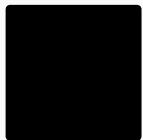
96 x 42 x 64



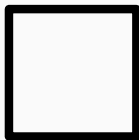
24 x 24 x 24

Evaluation Metrics

Block



Air



F1 Score

Input	Recon
	
	
	
	

False Negative

True Positive

True Negative

False Positive

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Evaluation Metrics

Reconstruction Loss

Token Embedding Vectors

0: air := [0.22, -1.22, 3.14, ...]

1: stair  := [1.02, -3.42, 0.02, ...]

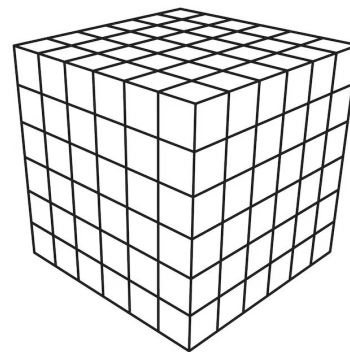
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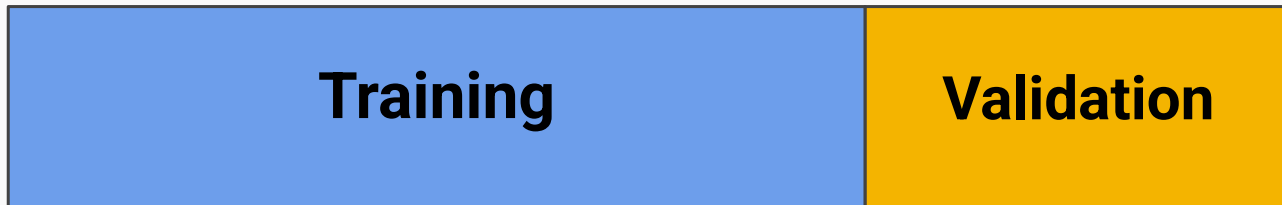
5: stone  := [1.10, 4.50, 1.12, ...]

(Per-voxel) Cosine Similarity of
Input & Reconstructed Vectors



Validation

Train / Validation Split



Languages & Packages



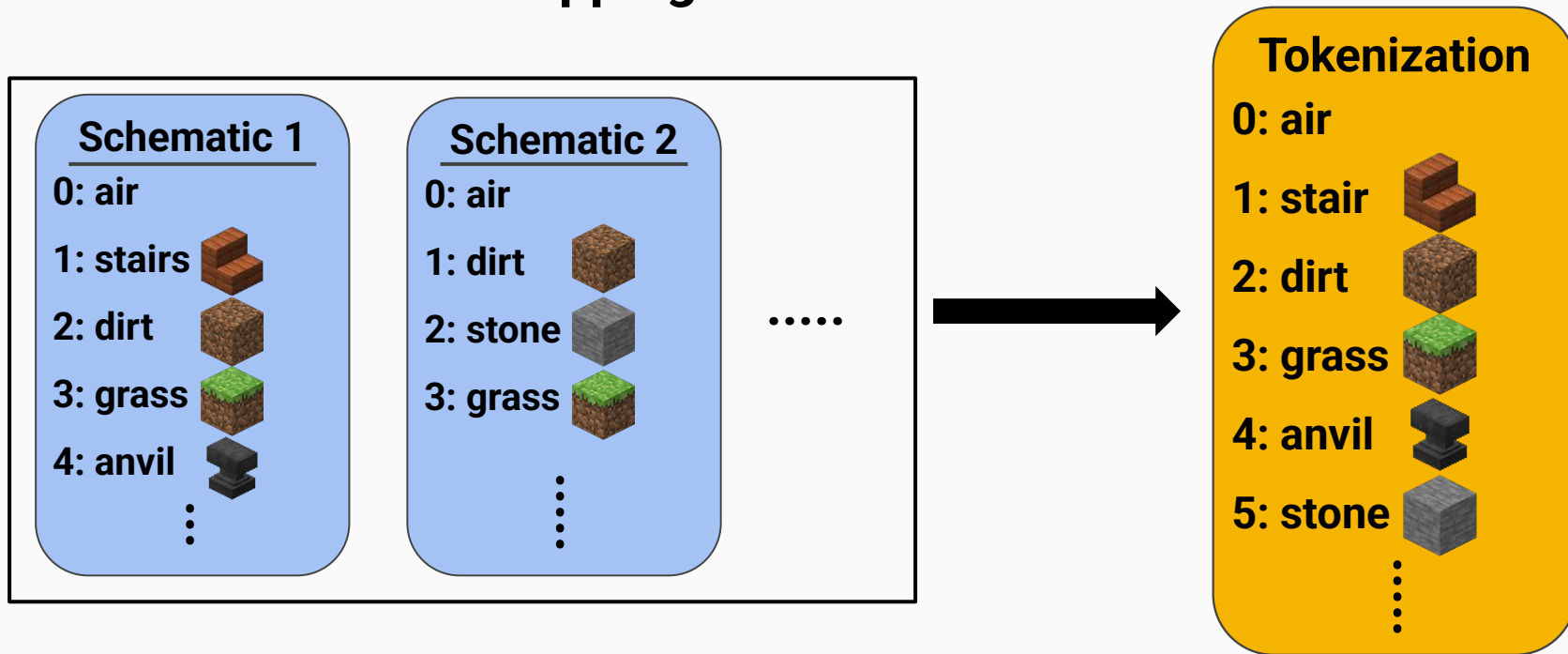
Python Packages for MC

- GDPC
- Amulet
- MCSchematic
- NBTLib

Software Design & Implementation

Dataset Pre-Processing

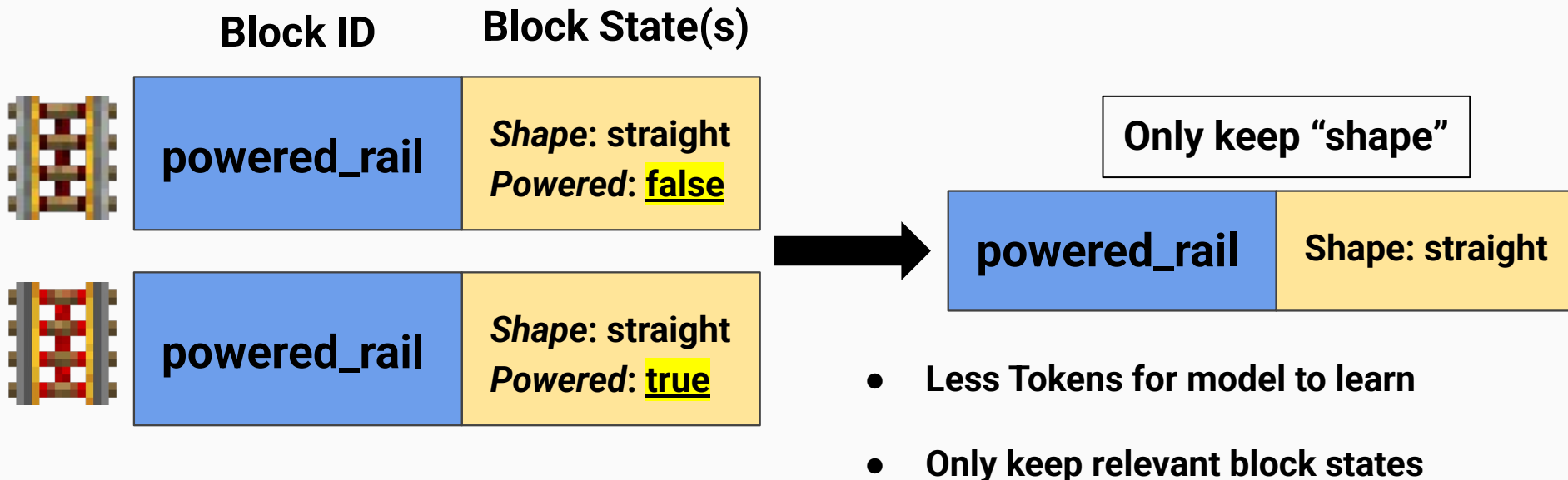
Block to Token Mapping



Software Design & Implementation

Dataset Pre-Processing

Blockstate Compression



Palette Class

Utility Class for managing Block to Token conversions (Tokenizations)

- Map to and from *Compressed Blockstate Tokenizations*
- Map Blocks to their transformed equivalent
- Mask Blocks to Air
- Validates the Tokenization:
 - Detects invalid block names
 - Fills in missing Blockstate combinations

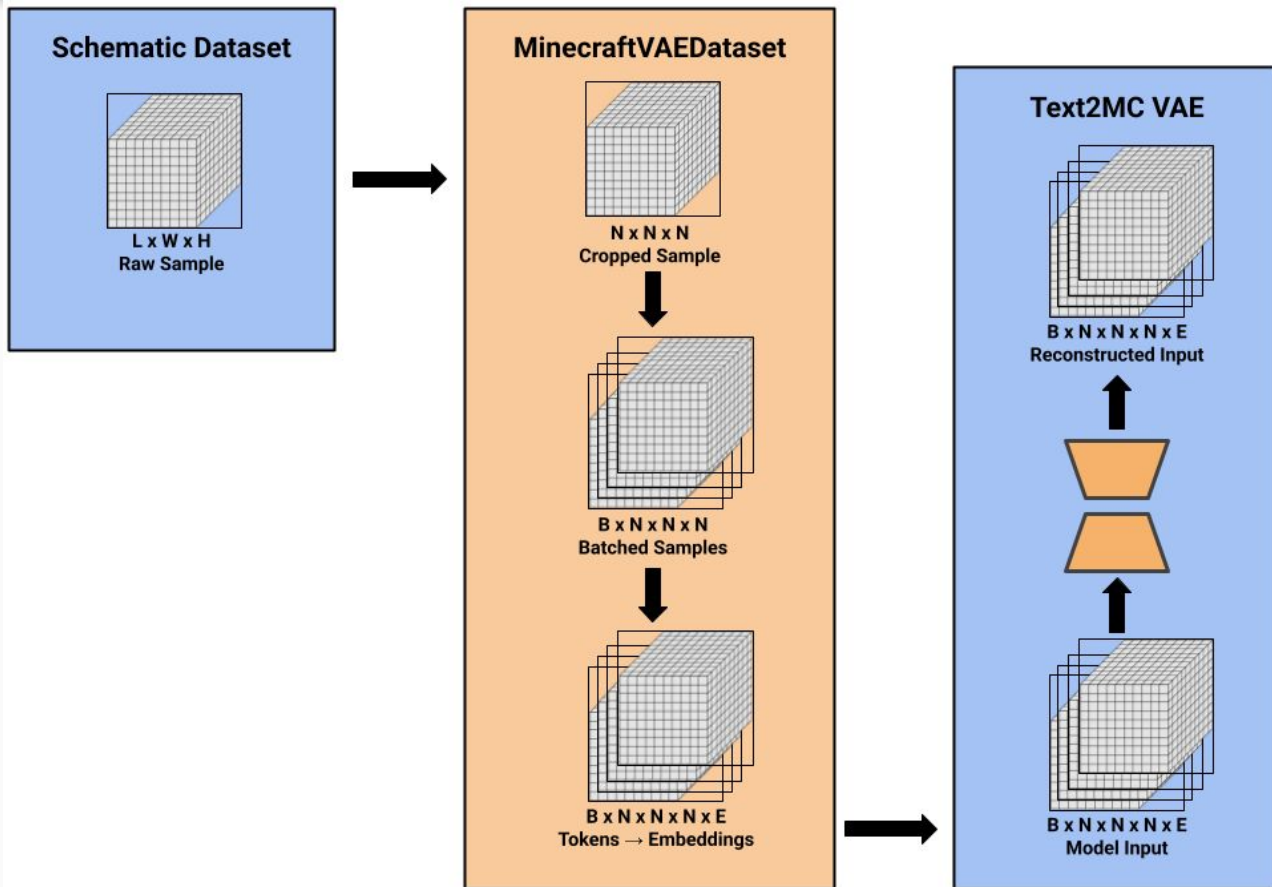


MinecraftVAEDataset Class

Implements PyTorch's Dataset Class

- Prepares Samples during training
- Implements Geometric Transforms & Weighted ROI Sampling
- Post-Processing Utility Functions for:
 - Converting Predicted Embeddings into Tokens
 - Saving Token arrays back into Schematic Files

Training Pipeline



Experiment Overview

Research Objective

Measure how Dataset Improvements Affect Reconstruction Quality to guide future research into Generative AI for Minecraft

Independent Variables:

1. Dataset Size
2. Random Geometric Transformations (True/False)
3. Weighted Random ROI Sampling (True/False)

Dependant Variables:

1. F1 Score
2. Reconstruction Loss

Experiment Design

Single Experiment Run

1. **Choose Configuration of Independent Variables**
2. **Retrain Text2MC VAE from scratch, only tuning Learning Rate**
3. **Train Until Validation Metrics stop improving**

Experiment Design

Training Configuration

Training Hyperparameters

(constant)

Sample Size	24 x 24 x 24
Batch Size	18
Precision	FP32
Model Architecture	Text2MC VAE
Model Scale	330M Parameters
Train / Val Split	90% / 10%

Experiment Run Configurations

(possible values)

Dataset Size	11K - 31K
Geometric Transforms	True/False
Weighted ROI Sampling	True/False
Learning Rate	1×10^{-7} - 1×10^{-5}

Results

Experiment Runs

Run #	Dataset Size	Geo Transforms	ROI Sampling	F1 Score	Recon Loss
Text2MC	11K	FALSE	FALSE	0.8540	0.3849
1	31K	TRUE	TRUE	0.8510	0.1153
2	22K	TRUE	TRUE	0.8497	0.1204
3	11K	TRUE	TRUE	0.8669	0.1690
4	11K	FALSE	TRUE	0.8229	0.1465

Example Reconstructions

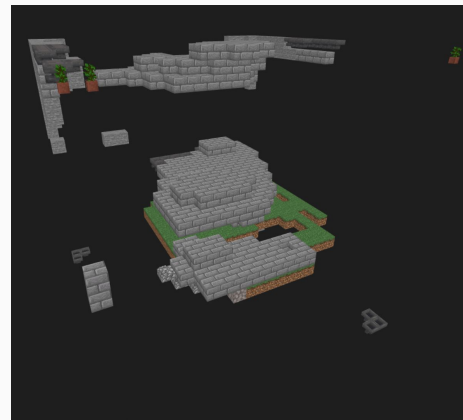
Original Sample



Run 3

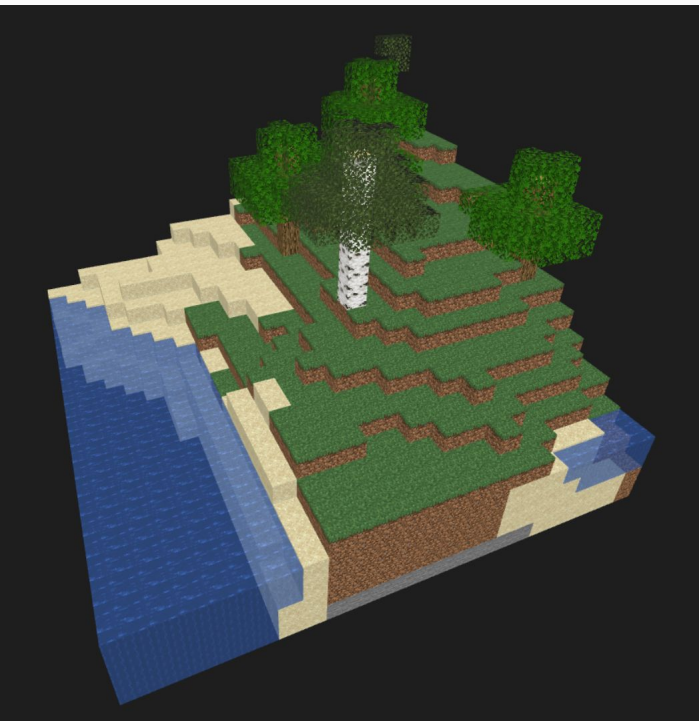


Text2MC

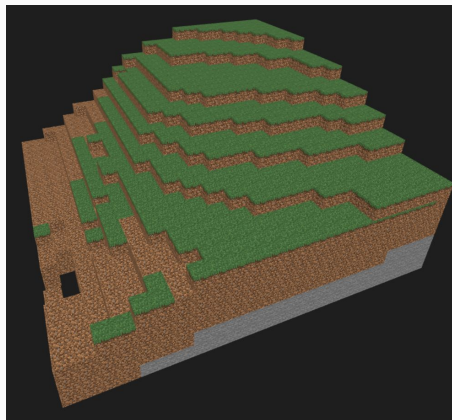


Example Reconstructions

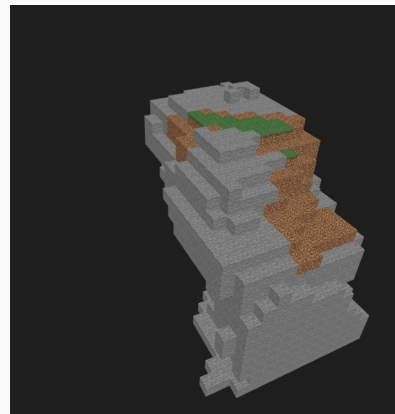
Original Sample



Run 3



Text2MC

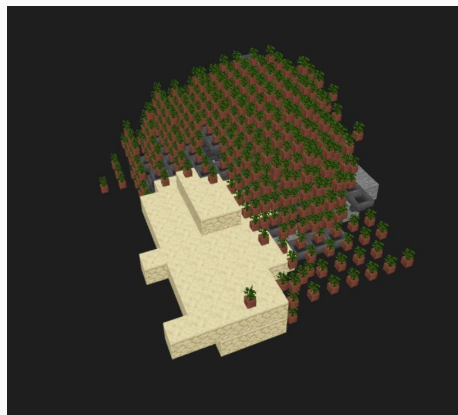


Example Reconstructions

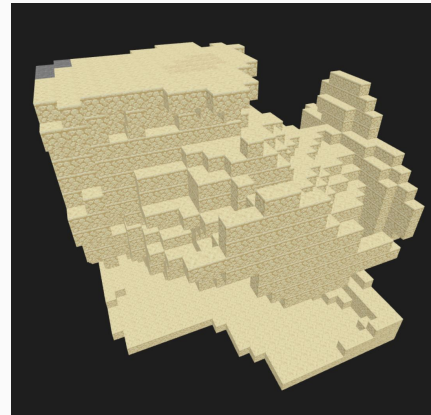
Original Sample



Run 3



Text2MC



Results Analysis and Observations

1. Dataset Size improves Reconstruction loss, little effect on F1.
2. Runs without Weighted ROI Sampling couldn't converge.
3. Geometric Transformations may improve both metrics, however more testing is needed.
 - Comparable Results to Text2MC on much less compute.
 - More Hyperparameter optimization could improve performance.

Code and Dataset Release

- **Data Processing & Training Code on Git**
- **Dataset Release on Kaggle**
- **Model Weights on HuggingFace**
- **ReadMe for Dataset & Code**
- **GDPC Package Contribution**
- **Share results in online forums**



Hugging Face

kaggle



git

Future Work

- **End-to-end Python Package for Minecraft & ML**
- **Process more available data**
- **Test different VAE Architectures**

Questions?

Key Points

- Machine Learning with Minecraft
- Related Work
- Dataset Analysis & Cleaning
- Experiment:
 - Research Question & Variables
 - Independent Variables
 - Eval Metrics - F1 & Cosine Similarity
 - Validation
 - Experiment Design
 - Experiment Runs & Results Analysis
- Code and Dataset Release
- Future Work